Circulation: Arrhythmia and Electrophysiology

ORIGINAL ARTICLE

Effect of Left Atrial Size and Volume Reduction on Pulmonary Vein Isolation Efficacy in Patients With Persistent and Longstanding Persistent Atrial Fibrillation: Subset Analysis From the aMAZE Trial

Dhanunjaya Lakkireddy, MD*; David J. Wilber, MD*; Suneet Mittal, MD*; David Rene Tschopp, MD; Christopher R. Ellis, MD; Abdi Rasekh, MD; Troy Hounshell, DO; Robert Hoyt, MD; Rudolph F. Evonich, MD; Sheetal Chandhok, MD; Ronald D. Berger, MD, PhD; Rodney Horton, MD; Michael Hoskins, MD; Hugh Calkins, MD; Steven J. Yakubov, MD; Omar K. Khalique, MD; Todd Cory Pulerwitz, MD; Nitish Badhwar, MD; Thomas Dewland, MD; Stanislav Weiner, MD; Smit Valsaiwala, MD; Aashish Katapadi, MD; Pamela Simons, BS, MBA; Patrick Walker, MPH; Nifi Ndilkintum, PhD; Luigi DiBiase, MD; Andrea Natale, MD; Randall J. Lee, MD, PhD; aMAZE Investigators†

BACKGROUND: Large left atrial volume (LAV) is a strong predictor of advanced left atrial substrate and is associated with less successful outcomes following pulmonary vein isolation (PVI) alone. Prespecified variables of the aMAZE trial (Left Atrial Appendage Ligation Adjunctive to PVI for Persistent or Longstanding Persistent Atrial Fibrillation) were evaluated to access for the impact of LAV and left atrial appendage ligation on atrial fibrillation (AF) outcomes.

METHODS: The aMAZE trial was a multicenter, randomized-controlled study evaluating the effects of LAV on freedom from atrial arrhythmias (AA) following PVI-only compared with LARIAT and PVI (LARIAT+PVI). In total, 610 drug-refractory patients with nonparoxysmal AF were randomized 2:1 to LARIAT versus PVI alone. Freedom from AA was assessed 12 months postprocedure. LAV was independently assessed by a core laboratory from cardiac computed tomography performed before ablation. (REGISTRATION: URL: https://www.clinicaltrials.gov; Unique identifier: NCT02513797)

RESULTS: There were 404 patients in the LARIAT+PVI group and 206 in the PVI-only group. Logistic regression performed within each subgroup for primary effectiveness with LAV demonstrated that the recurrence of AA after PVI was directly related to increasing LAV. Freedom from recurrence of AA in the LARIAT+PVI arm was independent and preserved irrespective of LAV. Similar results were seen with LAV index. A tercile analysis of early persistent AF (perAF) patients (AF >7 days and <6 months) and LARIAT+PVI in the highest (>148 cm^3) LAV tercile showed statistically significant freedom from AA compared with PVI-only (69% versus 49%; P=0.02). Significant differences (P<0.04) between groups in the early perAF cohort began at an LAV of 130 cm^3 and an LAV index of 65 cm^3/m^2.

CONCLUSIONS: The correlation of LAV with PVI effectiveness in this study demonstrates that AA recurrence after PVI alone is directly related to increasing LAV. Freedom from AA after left atrial appendage ligation in addition to PVI is independent of increasing LAV, and left atrial appendage ligation may therefore provide an adjunctive benefit in patients with nonparoxysmal AF and enlarged LAV.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: atrial fibrillation burden ■ left atrial appendage ligation ■ left atrial volume

Correspondence to: Dhanunjaya R. Lakkireddy, MD, The Kansas City Heart Rhythm, Institute (KCHRI) @ HCA Midwest, University of Missouri, Columbia, 12200, W 106th St, Overland Park, KS 66215, Email dlakkireddy@kchri.org; or Randall J. Lee, MD, PhD, University of California, San Francisco, 500 Parnassus Ave, San Francisco, CA 94143-1354, Email randall.lee@ucsf.edu

This manuscript was sent to Andrew E. Epstein, MD, Guest Editor, for review by expert referees, editorial decision, and final disposition.

*D.R. Lakkireddy, D.J. Wilber, and S. Mittal contributed equally.

†A list of all aMAZE investigators is given in the Supplemental Material.

Circulation: Arrhythmia and Electrophysiology is available at www.ahajournals.org/journal/circep

WHAT IS KNOWN?

- An enlarged left atrium reflects remodeling and is associated not only with an increased initial prevalence of atrial fibrillation but also with recurrence of arrhythmia after catheter ablation.
- Patients with an enlarged left atrium often require additional ablation strategies, and a few small studies suggest that left atrial appendage closure may have a role, but randomized clinical data are limited.

WHAT THE STUDY ADDS

- In this post hoc analysis of the aMAZE trial (Left Atrial Appendage Ligation Adjunctive to Pulmonary Vein Isolation for Persistent or Longstanding Persistent Atrial Fibrillation), we show that left atrial appendage ligation, performed adjunctively to pulmonary vein isolation, not only reduces left atrial size but also improves freedom from recurrent atrial arrhythmias in patients with nonparoxysmal atrial fibrillation.
- Our study suggests that adjunctive left atrial appendage ligation may be a strategy for treating atrial arrhythmias in patients with enlarged left atria.

Nonstandard Abbreviations and Acronyms

AA	atrial arrhythmias
AF	atrial fibrillation
LA	left atrial
LAA	left atrial appendage
LAV	left atrial volume
LAVI	left atrial volume index
NPAF	nonparoxysmal atrial fibrillation
PVI	pulmonary vein isolation

Catheter ablation is an established treatment for patients with symptomatic paroxysmal atrial fibrillation (AF).[1] Rhythm control after catheter ablation in patients with paroxysmal AF has been consistently better than that in those with nonparoxysmal AF (NPAF), who have a significantly higher recurrence rate. Various adjunctive strategies in the latter population have not been proven to improve ablation efficacy. Furthermore, left atrial (LA) remodeling is often associated with electrical remodeling, including areas of slow conduction, shortening of atrial refractoriness, and increased nonuniform anisotropy that facilitate reentrant circuits and AF.[2] LA enlargement reflects LA remodeling and is associated with an increased prevalence of AF.[3–7] An enlarged LA is also a well-recognized risk factor for AF recurrence after catheter ablation and is an important contributor, in addition to AF type, to predicting long-term catheter ablation success.[8–13] Small observational and randomized trials suggested that electrical isolation of the LA appendage (LAA) could lead to an adjunctive improvement in catheter ablation outcomes in patients with NPAF.[14,15] These patients commonly have a significantly enlarged LA compared with those with paroxysmal AF. However, data on the impact of LAA exclusion on enlarged LA size and on the outcomes of catheter ablation are unclear.

The aMAZE trial (LAA Ligation Adjunctive to Pulmonary Vein Isolation for Persistent or Longstanding Persistent Atrial Fibrillation) was a prospective, multicenter, externally monitored, randomized-controlled trial that tested the hypothesis that LAA ligation with the LARIAT LAA exclusion system (LARIAT; AtriCure, Inc, Mason, OH) as adjunctive therapy to pulmonary vein isolation (PVI) would lead to greater freedom from atrial arrhythmias (AA) compared with PVI alone in patients with nonparoxysmal AF undergoing initial catheter ablation (NCT02513797).[16] The coprimary end points were 30-day safety of the LARIAT procedure and freedom from documented AA of more than 30 seconds at 12 months after the initial PVI. A secondary outcome evaluating the technical success of LAA closure was also assessed. The primary aMAZE results were previously published and demonstrated that the primary safety end point was met, with a nonsignificant trend of a 4.3% increase in freedom from AA in the LARIAT plus PVI arm, and complete LAA closure in 84% of patients at 1 year, with 99% closure of <5-mm leaks.[17] Among the prespecified baseline variables that might influence the primary effectiveness end point were LA size and AF duration. Given the potential procedural risks and costs of catheter ablation, understanding the effects of variables that might influence its benefits and risks would help identify patients who might benefit most from catheter ablation alone or with LAA exclusion. Subsequently, this substudy analyzed the effects of LA size and AF duration from the aMAZE trial cohort on freedom from AA following PVI in patients with NPAF and compared the relationship between LA size and post-PVI AF recurrence with AF recurrence following LAA ligation plus PVI.

METHODS

Study Design

The study is a post hoc analysis of prespecified variables of the aMAZE trial (NCT02513797). The prespecified variables, LA size and AF duration, were analyzed in this post hoc analysis. The rationale, design, and protocol for the trial have been previously published.[16] Notably, the study received institutional review board approval at all participating sites and adhered to the guidelines set forth in the Declaration of Helsinki. The data that support the findings of this study are available from the corresponding author upon reasonable request.

Study Enrollment

All patients had documented symptomatic NPAF (7 days to 3 years of continuous AF), had failed at least 1 prior class I or III antiarrhythmic drug, and were eligible and planned to undergo catheter ablation. These patients were divided into early persistent AF (7 days to 6 months), late persistent AF (6–12 months), and longstanding persistent AF (12 months to 3 years). Key exclusion criteria included a LA diameter >6 cm, New York Heart Association Class IV heart failure, body mass index >40 kg/m^2, any prior procedure involving opening or entry into the pericardial space, and a documented thromboembolic event, myocardial infarction, or unstable angina within 3 months before potential enrollment. Additional details regarding the inclusion and exclusion criteria have been previously published.[17]

Following the initial screening visit and consent, all potentially eligible patients underwent confirmation of NPAF within 3 months of enrollment, a baseline cardiac computed tomography scan, and final adjudication of eligibility. Eligible patients were then consented for study enrollment and randomized to the treatment group (LARIAT+PVI) or control group (PVI-only) in a 2:1 fashion. The randomization schedule, as previously described, was masked to study patients, site personnel, and the sponsor.[16] Analysis was performed on a modified intention-to-treat population.

Study Oversight

The coprincipal investigators and executive committee had oversight of the study, with independent core laboratories validating rhythm monitoring outcomes and computed tomography/transesophageal echocardiography imaging data. A Clinical Events Committee and Data Monitoring Committee independently adjudicated safety events as well as monitored safety and performance end points and overall study integrity. An external clinical research organization (Avania, Inc, Marlborough, MA) operated and maintained the database, performed data analysis, and coordinated Clinical Events and Data Monitoring Committee activities. The study sponsor, AtriCure, Inc, remained blinded to the aggregate data and results throughout the trial.

LA Volume Calculation

LA volume (LAV) was determined by the core computed tomography laboratory in a blinded fashion. LAV was calculated using the biplane area-length method from reconstructed 4-chamber and 2-chamber views of the heart using the following formula, with *A* representing the area of the LA in the 4- or 2-chamber view, and *L* representing the shortest length from the center of the mitral annulus and perpendicular to the mitral annulus, intersecting the base of the LA within the 2 views.

$$A_L = \frac{8}{3\pi}\frac{A4c \cdot A2c}{L}$$

LAV index (LAVI) was calculated by dividing LAV by body mass index.

Statistical Analysis

A series of simple logistic regression models were conducted using the total randomization assignments in the aMAZE trial population, and separate individual logistic regression models were conducted for each treatment group.[18] Logistic regression analyses were used to examine the relation between freedom from AA and LAV and LAVI in the PVI-only and LARIAT+PVI groups. The logistic regression models were adjusted for age, sex, body mass index, and New York Heart Association heart failure class. All odds ratios were reported with their 95% CIs.

By using the full aMAZE sample size for the logistic regression analysis, the statistically ability to detect whether LAV is a general predictor of failure across both groups was improved and it directly tested whether the treatment effect (reduction in recurrence) varies depending on the baseline LAV or LAVI. It also relies on the randomized nature of the trial to ensure that other confounding variables are balanced, which is crucial when examining a variable like LAV or LAVI. The logistic regression analysis was performed on the actual success data from the aMAZE trial to determine the relationship between freedom from AA and LAV or LAVI. Model-based calculations based on the logistic regression were used to predict success at specific LAV values.

To corroborate the findings of the logistic regression analysis of the total aMAZE population, the PVI-only and LARIAT+PVI groups were treated as independent cohorts, with separate logistic analyses performed for each group. This allowed for the determination of whether a large LAV or LAVI affected ablation efficacy in the PVI-only group versus the LARIAT+PVI group.

This analysis was conducted in the early persistent AF population, the total persistent AF population, and the total persistent and longstanding persistent AF populations. Subgroup analyses based on LAV and LAVI were conducted by dividing subjects into terciles for each parameter. The χ^2 test of independence was used to compare the PVI-only versus LARIAT+PVI groups. A $P<0.05$ was considered statistically significant.

To further examine the impact of LAV on freedom from AA in the early persistent AF population, treatment groups were compared using subgroups of increasing LAV. This was done by starting with subjects with a volume of at least 115 mL and progressing to 175 mL in 5 cm^3 increments. Similarly, LAVI was evaluated from 57.5 mL/m^2 to 87.5 mL/m^2 in increments of 2.5 mL/m^2. A χ^2 test of independence was used to compare the primary success rate between the 2 treatment groups. A $P<0.05$ was again considered statistically significant.

RESULTS

Baseline Characteristics

In total, 610 patients from 53 institutions in the United States were enrolled and randomized to receive LARIAT+PVI (404 patients) or PVI-only (206 patients).[17] Patient characteristics were previously detailed and summarized (Table 1).[17]

Effects of LA Size: Logistic Regression Analysis

A logistic regression model based on the randomization assignment in the aMAZE trial was used to assess and predict the primary effectiveness of the treatment (LARIAT+PVI versus PVI-only) while controlling for LAV. A similar model was used while controlling for LAVI (Figure 1). All odds ratios were reported with their 95% CIs (Table 2).

Across the study population, in both the LARIAT+PVI and PVI-only treatment arms, the probability of freedom from AA decreased as LAV increased. In addition, across all populations, the LARIAT+PVI arm showed greater success than the PVI-only arm. For the early persistent population, the highest predicted success in the PVI-only arm

Table 1. Baseline Patient Characteristics

	LARIAT+PVI (n=404)	PVI (n=206)
Age, y		
Mean (±SD)	66.2 (8.4)	67.4 (7.5)
Median (Min–Max)	68.0 (29–80)	68.0 (40–80)
Sex, n (%)		
Male	288 (71)	158 (77)
Female	116 (29)	48 (23)
Body mass index, kg/m²		
Mean (±SD)	30.98 (4.6)	31.89 (4.5)
Median (Min–Max)	31.00 (19.2–42.6)	31.70 (20.7–41.0)
Medical history, n (%)*		
Hypertension, current or past	332 (82)	174 (84.5)
Hyperlipidemia, current or past, resolved, n (%)	242 (59.9)	122 (59.2)
Diabetes, current or past	75 (18.6)	48 (23.3)
Stroke, past, resolved	26 (6.4)	14 (6.8)
Transient ischemic attack or suspected neurological event, past, resolved	20 (5.0)	16 (7.8)
Thromboembolism, past, resolved, n (%)	17 (4.2)	7 (3.4)
Peripheral artery disease, current or past, resolved, n (%)	15 (3.7)	5 (2.4)
Smoking, current or former, n (%)	185 (46)	86 (42)
AF diagnosis, n (%)†		
Persistent	365 (88)	180 (87)
Longstanding persistent	48 (12)	26 (13)
Duration of AF, mo (n)		
Mean (±SD)	4.8 (8.3; 255)	4.6 (6.4; 135)
Median (Min–Max)	0.17 (0.0, 8.0)	0.17 (0.0, 3.0)
Duration of AF, mo; n/N (%)		
≥7 d to <6 mo	306/391 (78)	159/195 (82)
≥6 mo to <12 mo	39/391 (10)	15/195 (8)
≥12 mo	46/391 (12)	21/195 (11)
History of AAD use, n (%)‡		
Class IB	2 (<1)	0 (0)
Class IC	144 (36)	74 (36)
Class III	301 (75)	154 (75)
CHA_2DS_2-VASc score		
Mean (±SD)	1.6 (1.10)	1.6 (1.17)
Median (Min–Max)	1.0 (0, 5)	1.0 (0, 6)
NYHA classification, n (%)§	N=400	N=205
I	59 (15)	38 (18)
II	106 (26)	45 (22)
III	31 (8)	18 (9)
No history of heart failure	204 (50)	104 (50)
Left atrium diameter, mm (n)		
Mean (±SD)	44.11 (6.0–390)	45.38 (6.9–200)
Median (Min–Max)	44.10 (25.7–59.7)	45.60 (5.7–58.8)
Left atrial volume, mL (n)		
Mean (±SD)	135.31 (38.35–388)	141.92 (39.55–199)
Median (Min–Max)	131.75 (45.6–283.0)	137.90 (65.2–258.0)
Left atrial volume index, mL/m²		
Mean (±SD)	63.86 (18.1)	64.94 (18.0)
Median (Min–Max)	62.94 (21.5–138.0)	63.04 (25.5–120.9)

(*Continued*)

Table 1. Continued

AAD indicates antiarrhythmic drug; AF, atrial fibrillation; aMAZE, Left Atrial Appendage Ligation Adjunctive to Pulmonary Vein Isolation for Persistent or Longstanding Persistent Atrial Fibrillation; CHA_2DS_2-VASc score, congestive heart failure/LV dysfunction, hypertension, age ≥ 75 years old - diabetes, stroke/TIA/TE- vascular disease, age 65-74 years old, sex category; LA, left atrium; NYHA, New York Heart Association; and PVI, pulmonary vein isolation.

*Medical history was assessed by authorized staff at participating sites using a case report form.

†Persistent AF is defined as AF sustained for ≥7 d and ≤1 y; longstanding persistent AF is defined as continuous AF >1 y duration.

‡No subjects had a history of Class IA AAD use.

§NYHA Classes range from I to IV, defined as follows: NYHA I: no limitation of physical activity; ordinary physical activity does not cause undue fatigue, palpitation, or shortness of breath; II: Slight limitation of physical activity; comfortable at rest; ordinary physical activity results in fatigue, palpitation, shortness of breath. III: Marked limitation of physical activity; comfortable at rest; less than ordinary activity causes fatigue, palpitation, shortness of breath. The exhibition of NYHA IV symptoms was an exclusion criterium for aMAZE (unable to carry on any physical activity without discomfort; symptoms of heart failure at rest; any physical activity causes further discomfort).

was 63% (95% CI, 50%–75%) at an LAV of 65.2 mL. The corresponding predicted success at an LAV of 65.1 mL in the LARIAT+PVI arm was 70% (95% CI, 60%–79%). The predicted effectiveness was at least 60% for all LAVs ≤208.2 mL in the LARIAT+PVI arm, but was observed only at LAVs ≤118 mL in the PVI-only arm, representing < 25% of the population. Similar results were observed in the combined early and late persistent population, where the effective treatment with at least 60% success was observed in all patients in the LARIAT+PVI arm, but only in 25% or fewer of patients in the PVI-only arm. For the entire persistent and longstanding persistent population, the predicted effectiveness was at least 60% for all LAVs ≤172.6 mL in the LARIAT+PVI arm but was only seen at LAVs ≤152 mL in the PVI-only arm.

For further exploration, individual logistic regression models were analyzed for each treatment group, controlling for LAV and LAVI (Figure 2). Across the study populations, in the PVI-only arm, we observed a steady decline in predicted freedom from AA as LAV increased. In contrast, the LARIAT+PVI arm demonstrated a relatively constant predicted probability of freedom from AA of ≈65%.

Effect of LA Size: Tercile Analysis

Tercile analysis of freedom from AA in the early persistent AF population demonstrated that the highest

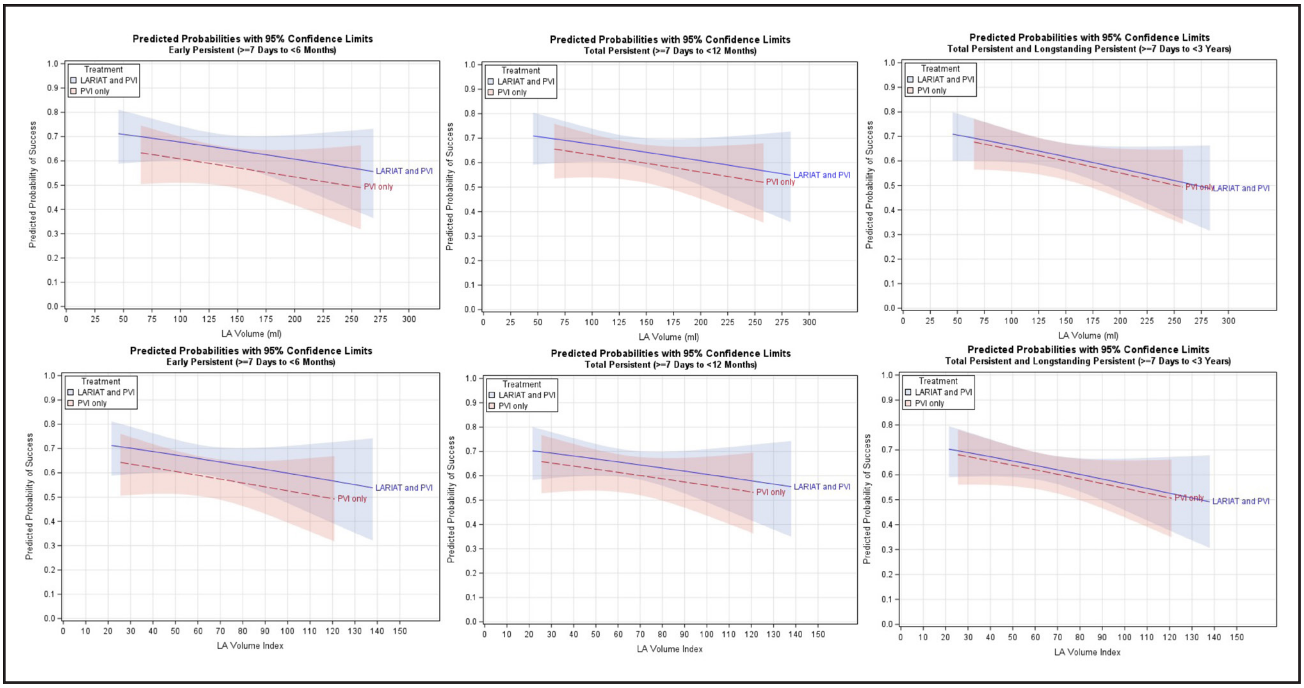

Figure 1. Predicted proportions of treatment efficacy.
Logistic regression analyses were performed on the full aMAZE (Left Atrial Appendage Ligation Adjunctive to Pulmonary Vein Isolation for Persistent or Longstanding Persistent Atrial Fibrillation) population. The efficacy of treatment (either pulmonary vein isolation [PVI]+LARIAT or PVI-only) was analyzed while controlling for left atrial (LA) volume and LA volume index.

LAV and LAVI terciles were associated with a statistically significant benefit in freedom from AA in the LARIAT+PVI group compared with the PVI-only group (P<0.035 and P<0.034), with a 19% absolute difference between the groups for both LAV and LAVI (Table 3). There was a similar trend in the highest LAV and LAVI terciles in the combined early and late persistent AF population, with an absolute difference of 16% between the LARIAT+PVI and PVI-only groups, but this difference was not statistically significant (P<0.055). In the highest LAV tercile, there was an absolute difference of 12% between the LARIAT+PVI and PVI-only groups in the total persistent and longstanding persistent population, which, again, was not statistically significant (P<0.143).

Threshold Analysis

In the early persistent AF population, a significant difference in freedom from AA between LARIAT+PVI and PVI-only for LAV and LAVI began at 130 mL and 65 mL/m^2, respectively. Although there were persistent absolute differences in freedom from AA between LARIAT+PVI and PVI-only for LAV (14%–21%) and LAVI (14%–27%), respectively, the statistically significant difference in freedom from AA between LARIAT+PVI and PVI-only persisted (Figure 3) for several increments in LAV (130–155 mL) and LAVI (65–77 mL/m^2).

DISCUSSION

This study demonstrates that LAA ligation combined with PVI provides a significant benefit over PVI alone in preventing atrial arrhythmia recurrence in NPAF patients with an enlarged LA. In contrast to the decreasing efficacy of PVI as LAV increases in the PVI-only group, LAA ligation in addition to PVI appears to be associated with higher success rates in preventing recurrence of AA and maintains a relatively more constant rate of the freedom from AA across a range of preprocedure LAVs.

This study is the first to demonstrate that LAA ligation in persistent AF with an enlarged LA is associated with a reduction in AF recurrence after PVI. LAA ligation creates ischemic necrosis of the LAA with atrophy and resorption of the LAA, analogous to excision of the LAA, thus reducing LA volume.[19,20] Computed tomography imaging pre- and post-LAA ligation has shown a significant decrease in LAV.[21] In addition, LAA ligation has been shown to electrically isolate the LAA distal to the suture ligature, produce spontaneous termination of AA arising from the body of the LAA, and produce favorable electrical remodeling.[22–26] Although freedom from AA end point was not met in the original aMAZE trial, implying that electrical isolation of the body of the LAA is not a generalizable adjunctive strategy to PVI, electrical isolation of the LAA ostium with catheter ablation has been shown to be effective in increasing the efficacy of

Table 2. Freedom From Atrial Arrhythmias Between the LARIAT Plus PVI and PVI-Only Treatment Groups

Population	Unadjusted, odds ratio (95% CI)*	LAV adjusted, odds ratio (95% CI)	LAVI adjusted, odds ratio (95% CI)
Early persistent (<7 d to 6 mo)	1.38 (0.90–2.11)	0.997 (0.992–1.002)	0.994 (0.982–1.005)
Persistent (<7 d to 12 mo)	1.24 (0.83–1.86)	0.997 (0.992–1.002)	0.995 (0.984–1.005)
aMAZE total population	1.13 (0.77–1.65)	0.996 (0.991–1.001)	0.992 (0.982–1.002)

*95% Wald confidence limits; odds ratio >1 shows benefit of Lariat compared with PVI-only; although not statistically significant, adjusting for LAV or LAVI in the logistic regression model altered the treatment effect estimate. aMAZE indicates Left Atrial Appendage Ligation Adjunctive to PVI for Persistent or Longstanding Persistent Atrial Fibrillation; LAV, left atrial volume; LAVI, left atrial volume index; and PVI, pulmonary vein isolation.

PVI in NPAF patients.[14,15,19,27] However, LAA reconnection is high with catheter ablation; LAA electrical isolation via catheter ablation can also result in a high incidence of LAA thrombus and an increased incidence of ischemic stroke.[28,29] Subsequently, LAA ligation has been shown to favorably remodel electrical activity beyond LAA isolation achieved by catheter ablation.[26]

NPAF patients have worse electroanatomic remodeling of the LA than patients with paroxysmal atrial fibrillation and thus have a large LAV. An increase in LA size has long been recognized as an independent risk factor for AF.[3–5] Population-based studies have demonstrated that the incidence of AF increases by 39% with each 5 mm increase in LA diameter.[4] Similarly, LAV is a robust independent predictor of increased LA size and AF recurrence.[8,9] These observations are consistent with the prior work of Garrey,[30] who in 1914 hypothesized that a critical mass of atrial tissue is necessary to sustain AF.[31] It is, thus, generally accepted that LA size matters in the cause and prevalence of AF, and it has been demonstrated that reduction of the critical electrical mass of the LA contributes to the effectiveness of AF ablation.[32,33] It is not surprising that patients with an enlarged LA do not have a sufficient reduction of critical electrical surface area by PVI alone, potentially explaining the poor freedom from AF following PVI in patients with NPAF, as several studies have suggested that catheter ablation reduces the effective LA surface area, which perpetuates AF and has been correlated with freedom from AF recurrence.[9,34–36]

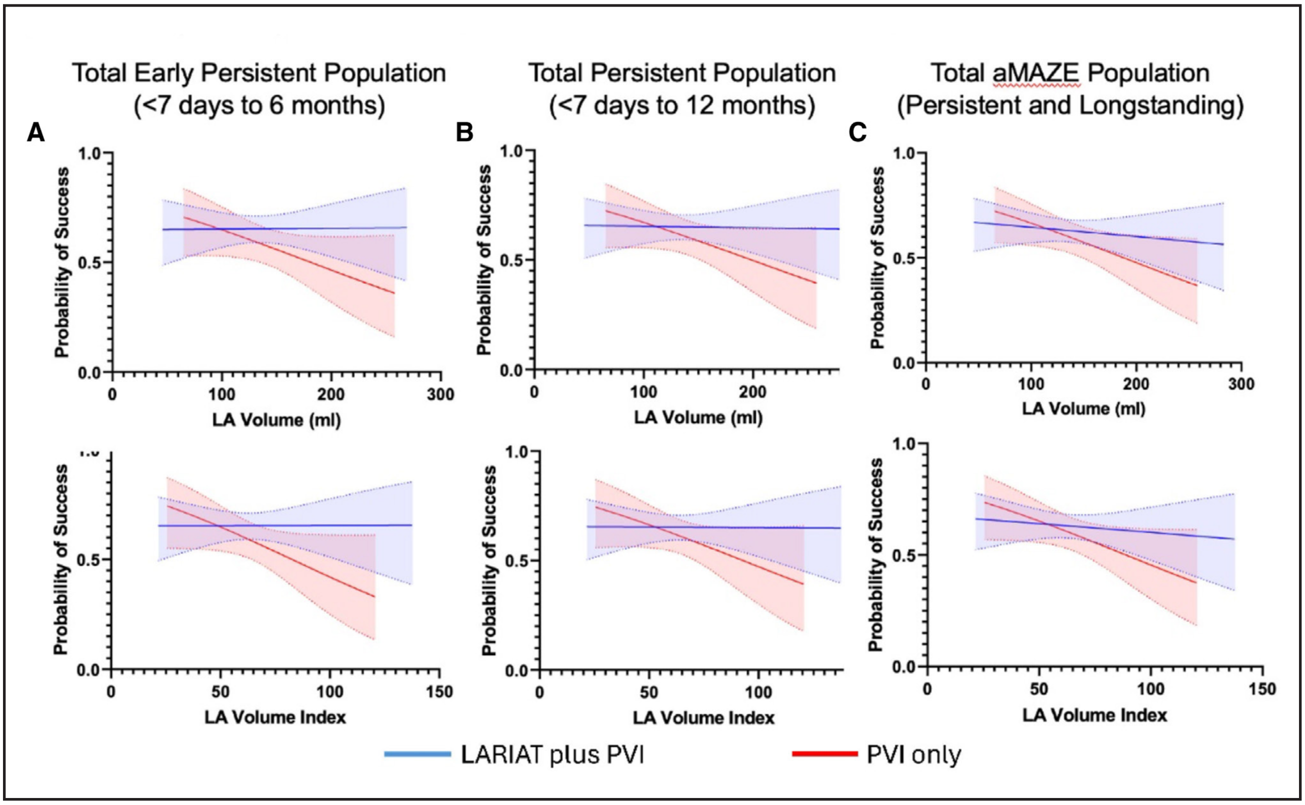

Figure 2. Predicted probability of treatment efficacy in pulmonary vein isolation (PVI)-only vs LARIAT+PVI.
Individual logistic regression analyses were performed on the PVI-only and LARIAT+PVI groups as separate cohort populations, controlling for left atrial (LA) volume and LAV index. Logistic analyses were performed separately on the PVI-only and LARIAT+PVI groups and then combined in this summary figure. PVI-only populations are represented as red lines. LARIAT+PVI is represented as blue lines. **A**, Total early persistent population. **B**, Total persistent population. **C**, Total aMAZE (Left Atrial Appendage Ligation Adjunctive to PVI for Persistent or Longstanding Persistent Atrial Fibrillation) population. Probability of success by LAV according to treatment arm.

Table 3. Tercile Analysis of the Primary Effectiveness End Point in the Early Persistent Group (>7 Days and <6 Months)

LAV					LAVI				
Primary success	LARIAT+PVI, n=286	PVI-only, n=153	Difference (95% CI)	*P* value*	Primary success	LARIAT+PVI, n=286	PVI-only, n=153	Difference (95% CI)	*P* value*
Lower tercile	65% (60/92)	61% (27/44)	4% (−13% to 21%)	0.66	Lower tercile	66% (56/85)	66%(31/47)	0% (−17% to 17%)	0.99
Medium tercile	64% (51/80)	65% (28/43)	−1% (−19% to 16%)	0.88	Medium tercile	61% (50/82)	58% (25/43)	3% (−15% to 21%)	0.76
High tercile	67% (47/70)	48% (25/52)	19% (2% to 37%)	0.03	High tercile	70% (53/76)	51% (25/49)	19% (1% to 36%)	0.04
Tercile 1 cutoff: 117.8; tercile 2 cutoff: 148.9 as defined in the ITT population. [1] Subjects included in analysis are those evaluable for primary efficacy end point.					Tercile 1 cutoff: 54.9; tercile 2 cutoff: 69.7 as defined in the analysis population. [1] Subjects included in analysis are those evaluable for primary efficacy end point.				

*P value from χ^2 test of independence was used to compare the primary success rate between the 2 treatment groups. ITT indicates intention to treat; LAV, left atrial volume, LAVI, left atrial volume index; and PVI, pulmonary vein isolation.

In addition to the observed antiarrhythmic effect of LAA ligation, there is also the potential benefit of eliminating a nidus for LAA thrombus formation with LAA ligation. Although the aMAZE trial was not a preventive stroke study, the initial aMAZE results reported transesophageal echocardiography-confirmed complete closure rates with no leaks of 84% at 12 months; 99% of patients had leaks < 5 mm, if present, at 12 months. The aMAZE trial results are consistent with previous observational and registry data for LAA ligation closure, which also reported stroke rates post-LAA ligation without oral anticoagulation therapy in the 1% to 2% range.[37–39] Leaks in partially atrophied, electrically inactive LAAs could increase the risk of thromboembolic events, but large registries have not demonstrated this association. Because the leak following LAA ligation is a central leak, it can be easily closed with a vascular plug with 100% success.[40]

The PVI-only arm of the aMAZE trial highlights the impact of an enlarged LA within the total population of persistent and longstanding persistent AF. As the LAV increased, the efficacy of freedom from AA decreased with LAV >130 cm^3. This was associated with a <53% freedom from AA, which continued to decline with increasing LAV. Only 25% of the early persistent AF population with LAV <115 cm^3 was able to achieve at least 60% freedom from AA. Similar results were observed in the total persistent and longstanding persistent PVI-only population, where an efficacy rate of at least 60% was achieved in < 25% of the population. These results are similar to those of the recently reported DECAF II substudy (Determining the Association of Chromosomal Variants with Non-PV Triggers and Ablation-Outcome in AF), in which LAV was found to be the strongest predictor of atrial arrhythmia recurrence after PVI in patients with persistent AF.[41,42] Patients with a LAV <114 mL had a 73% 1-year freedom from atrial arrhythmia recurrence, whereas patients with a LAV >114 mL had <53% freedom from recurrence, a difference that worsened as LAV increased.[42] In patients with recurrent AF despite durable PVI, various ablation strategies used alone or in combination during the redo procedure have not improved arrhythmia-free survival.[13] These contemporary, large studies in NPAF and redo ablation in patients with durable PVI indicate that the likelihood of recurrent AF is high with an enlarged LA. Thus, performing PVI-only ablation should be weighed against the costs and risks of catheter ablation.

Although LAA ligation did not provide statistically significant improvement in atrial arrhythmia recurrence

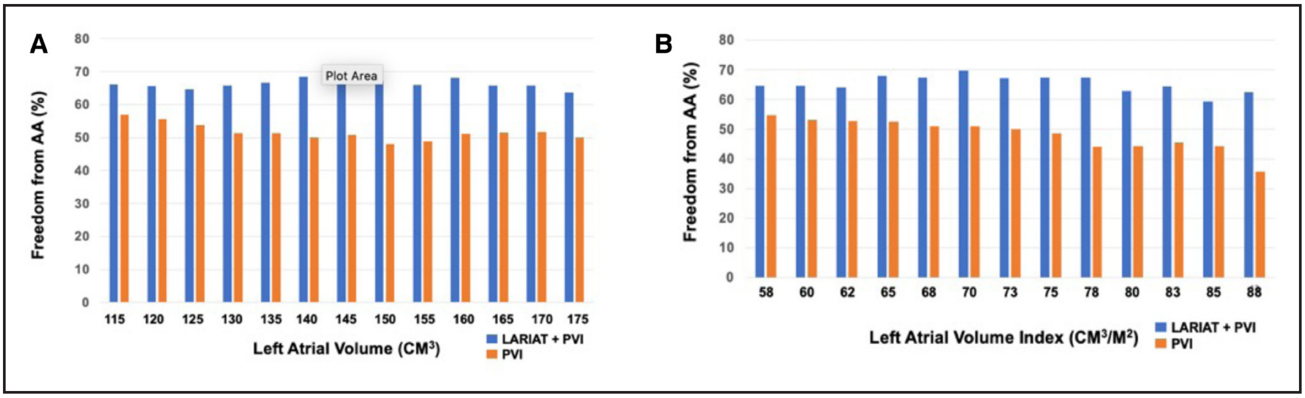

Figure 3. Left atrial volume and index threshold analyses in early persistent atrial fibrillation (>7 days and <6 months).
A, Freedom from atrial arrhythmias (AA) by left atrial volume. **B**, Freedom from AA by left atrial volume index, by treatment arm. PVI indicates pulmonary vein isolation.

across all NPAF patients in the aMAZE study, its demonstrable impact in preserving rhythm control in patients with significant LA enlargement is worth highlighting. Mechanistically, reduction in available atrial substrate for AF perpetuation may play a bigger role than the electrical triggers originating from the body of the LAA in those patients with significant atrial remodeling with associated LA enlargement >130 cm^3. Since LAA ligation does not generally affect the LAA ostium, where the majority of LAA triggers originate, LAA ligation has a different mechanism than ablation of the LAA base and may have additive beneficial effects.[14,15] Perhaps a more detailed understanding of the atrial electrical substrate and LAV interaction with PVI and LAA ligation should be further studied.

Limitations

The aMAZE population was not an evenly distributed cohort among the 3 AF classification groups of early persistent (7 days to 6 months), late persistent (6–12 months), and longstanding persistent (12 months to 3 years) AF. Most of the aMAZE population consisted of the early persistent group (79% of the overall population). This distribution may be due to investigator bias that more advanced NPAF depends on nonpulmonary vein substrate, such as the posterior LA wall, and needs additional ablation beyond PVI, thus leading to decreased enrollment in the late- and longstanding-persistent AF groups. Nevertheless, the beneficial effects of LAA ligation and PVI were also observed in the late- and longstanding-persistent groups. However, statistical significance was not reached, potentially due to the small number of patients in each group and the better-than-expected efficacy rate of the PVI-only group in the longstanding persistent AF population. An alternative explanation is that, in late- and longstanding-persistent AF patients, LA posterior wall isolation is also important.[43–45] Since additional substrate changes associated with atrial myopathy, such as fibrosis, were not systematically assessed, this may have influenced outcomes. Recent randomized trials have also demonstrated the benefit of combining LAA epicardial exclusion and LA posterior wall isolation.[46,47] In addition, the study population excluded patients with an LA diameter >6 cm, with the largest LAV being 208 mL. Therefore, the potential benefit of LAA ligation plus PVI may only be applicable to patients with an LA diameter <6 cm or LAV <208 mL (which was the upper range of LAV observed). Lastly, potential confounding factors could be related to baseline characteristic imbalances in the tercile analysis. However, comparing baseline characteristics within terciles is prone to the chance of imbalance because of the reduced sample size in each of the subgroups. Despite these limitations, our results suggest that LAA ligation in addition to PVI may be helpful in patients with an enlarged LA, but additional trials are still needed.

Conclusions

Recurrence of AA following PVI significantly increases as LAV increases. LAA ligation plus PVI is associated with superior freedom from AA compared with PVI alone in early persistent AF patients with enlarged LAV, suggesting that LAA ligation in patients with enlarged LAV may be a viable adjunctive therapy to PVI. Future studies are needed to determine whether there is an optimal ablation lesion set combined with LAA ligation that provides an adjunctive benefit to PVI alone for a broader population of patients with NPAF with enlarged LA.

ARTICLE INFORMATION

Received December 9, 2025; accepted June 30, 2026.

Affiliations

Kansas City Heart Rhythm Institute, Overland Park (D.L., A.K.). Loyola University Chicago Stritch School of Medicine, Maywood, IL (D.J.W., S.V.). Valley Health System, Ridgewood, NJ (S.M.). Texas Cardiac Arrhythmia Institute, Austin (D.R.T., R. Horton, A.N.). Vanderbilt University Heart-EP, Nashville, TN (C.R.E.). Baylor College of Medicine, Houston, TX (A.R.). MercyOne Iowa Heart Center, West Des Moines, IA (T.H., R. Hoyt). Upper Peninsula Health System-Marquette, MI (R.F.E.). Bryn Mawr Medical Specialists Association, PA (S.C.). Johns Hopkins University, Baltimore, MD (R.D.B., H.C.). New Mexico Heart Institute, Albuquerque (M.H.). Ohio Health Heart & Vascular, Columbus (S.J.Y.). St. Francis Hospital & Heart Center, Roslyn, NY (O.K.K.). Columbia University Irving Medical Center, New York, NY (T.C.P.). Stanford University, CA (N.B.). University of California San Francisco, San Francisco (T.D., R.J.L.). Christus Health, Tyler, TX (S.W.). AtriCure, Inc., Mason, OH (P.S., N.N., R.J.L.). Avania, Marlborough, MA (P.W.). Albert Einstein College of Medicine at Montefiore Health System, Bronx, NY (L.D.B.). University of Tor Vergata, Rome, Italy (A.N.). Case Western Reserve University, Cleveland, OH (A.N.).

Acknowledgments

Marcie Meador, PhD (AtriCure, Inc), and Kristen Plasseraud, PhD (AtriCure, Inc), provided medical writing and editing assistance under the direction of the investigators.

Sources of Funding

The study was funded by SentriHeart/AtriCure, Inc, which remained blinded to the aggregate data and results throughout the trial.

Disclosures

Dr Lakkireddy is a consultant to Abbott, Acutus, AltaThera, Atricure, Biotronik, Boston Scientific, Biosense Webster, Lifetech, Medtronic, Philips, and Sanofi. Dr Wilber reports grants from AtriCure (research grant to institution for conduct of the aMAZE trial; no personal fees or salary) and grants from AtriCure (research grant for conduct of the aMAZE trial (Left Atrial Appendage Ligation Adjunctive to Pulmonary Vein Isolation for Persistent or Longstanding Persistent Atrial Fibrillation) to institution; no personal fees or salary) during the conduct of the study; grants from Biosense Webster (research grant to institution), grants from Abbott (research grant to institution), grants from Boston Scientific (research grant to institution), personal fees from Biosense Webster (educational lectures, advisory board), and personal fees from Medtronic (educational lectures) outside the submitted work. Dr Mittel reports receiving consulting fees or honoraria from Boston Scientific, Medtronic, Philips, SentreHeart, Abbott, and AtriCure. Dr Mittal also reports serving on a data and safety monitoring board/advisory committee for Catawba and Boston Scientific. Dr Ellis has received research grants to Vanderbilt University Medical Center from Atricure, Inc, Boston Scientific, Inc, and Medtronic, Inc. Dr Ellis has received consulting or advisory fees from Abbott Medical, Inc, Atricure, Inc, Boston Scientific, Inc, and Medtronic, Inc. P. Simons reports receiving salary support as an employee of AtriCure, Inc. Dr Yakobov receives honoraria from AtriCure, Medtronic, and Boston Scientific. P. Walker is employed by Avania, Inc, which received payment to perform the statistical analysis. Dr Lee is a part-time medical director at AtriCure. The other authors report no conflicts.

Supplemental Material

aMAZE Investigators

REFERENCES

1. Joglar JA, Chung MK, Armbruster AL, Benjamin EJ, Chyou JY, Cronin EM, Deswal A, Eckhardt LL, Goldberger ZD, Gopinathannair R, et al. 2023 ACC/AHA/ACCP/HRS guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2023;83:109. doi: 10.1016/j.jacc.2023.08.017
2. Delgado V, Di Biase L, Leung M, Romero J, Tops LF, Casadei B, Marrouche N, Bax JJ. Structure and function of the left atrium and left atrial appendage: AF and stroke implications. *J Am Coll Cardiol*. 2017;70:3157–3172. doi: 10.1016/j.jacc.2017.10.063
3. Vaziri SM, Larson MG, Benjamin EJ, Levy D. Echocardiographic predictors of nonrheumatic atrial fibrillation. The Framingham Heart Study. *Circulation*. 1994;89:724–730. doi: 10.1161/01.cir.89.2.724
4. Psaty BM, Smith NL, Siscovick DS, Koepsell TD, Weiss NS, Heckbert SR, Lemaitre RN, Wagner EH, Furberg CD. Health outcomes associated with antihypertensive therapies used as first-line agents: a systematic review and meta-analysis. *JAMA*. 1997;277:739–745. doi: 10.1001/jama.1997.03540330061036
5. Tsang TS, Barnes ME, Bailey KR, Leibson CL, Montgomery SC, Takemoto Y, Diamond PM, Marra MA, Gersh BJ, Wiebers DO, et al. Left atrial volume: important risk marker of incident atrial fibrillation in 1655 older men and women. *Mayo Clin Proc*. 2001;76:467–475. doi: 10.4065/76.5.467
6. Pritchett AM, Jacobsen SJ, Mahoney DW, Rodeheffer RJ, Bailey KR, Redfield MM. Left atrial volume as an index of left atrial size: a population-based study. *J Am Coll Cardiol*. 2003;41:1036–1043. doi: 10.1016/s0735-1097(02)02981-9
7. Gupta DK, Shah AM, Giugliano RP, Ruff CT, Antman EM, Grip LT, Deenadayalu N, Hoffman E, Patel I, Shi M, et al; Effective aNticoaGulation with factor xA next GEneration in AF-Thrombolysis In Myocardial Infarction 48 Echocardiographic Study Investigators. Left atrial structure and function in atrial fibrillation: ENGAGE AF-TIMI 48. *Eur Heart J*. 2014;35:1457–1465. doi: 10.1093/eurheartj/eht500
8. Abecasis J, Dourado R, Ferreira A, Saraiva C, Cavaco D, Santos KR, Morgado FB, Adragão P, Silva A. Left atrial volume calculated by multidetector computed tomography may predict successful pulmonary vein isolation in catheter ablation of atrial fibrillation. *Europace*. 2009;11:1289–1294. doi: 10.1093/europace/eup198
9. Costa FM, Ferreira AM, Oliveira S, Santos PG, Durazzo A, Carmo P, Santos KR, Cavaco D, Parreira L, Morgado F, et al. Left atrial volume is more important than the type of atrial fibrillation in predicting the long-term success of catheter ablation. *Int J Cardiol*. 2015;184:56–61. doi: 10.1016/j.ijcard.2015.01.060
10. Njoku A, Kannabhiran M, Arora R, Reddy P, Gopinathannair R, Lakkireddy D, Dominic P. Left atrial volume predicts atrial fibrillation recurrence after radiofrequency ablation: a meta-analysis. *Europace*. 2018;20:33–42. doi: 10.1093/europace/eux013
11. Strisciuglio T, Di Gioia G, Chatzikyriakou S, Silva Garcia E, Barbato E, Geelen P, De Potter T. Left atrial volume computed by 3D rotational angiography best predicts atrial fibrillation recurrence after circumferential pulmonary vein isolation. *Int J Cardiovasc Imaging*. 2018;34:337–342. doi: 10.1007/s10554-017-1243-7
12. Hwang J, Park HS, Han S, Jun SW, Kang NY, Jeon JH, Choi SW, Lee CH, Kim IC, Cho YK, et al. The impact of catheter ablation of atrial fibrillation on the left atrial volume and function: study using three-dimensional echocardiography. *J Interv Card Electrophysiol*. 2020;57:87–95. doi: 10.1007/s10840-019-00696-8
13. Benali K, Barré V, Hermida A, Galand V, Milhem A, Philibert S, Boveda S, Bars C, Anselme F, Maille B, et al. Recurrences of atrial fibrillation despite durable pulmonary vein isolation: the PARTY-PVI study. *Circ Arrhythm Electrophysiol*. 2023;16:e011354. doi: 10.1161/CIRCEP.122.011354
14. Di Biase L, Burkhardt JD, Mohanty P, Mohanty S, Sanchez JE, Trivedi C, Gunes M, Gokoglan Y, Gianni C, Horton RP, et al. Left atrial appendage isolation in patients with longstanding persistent AF undergoing catheter ablation: BELIEF trial. *J Am Coll Cardiol*. 2016;68:1929–1940. doi: 10.1016/j.jacc.2016.07.770
15. Friedman DJ, Black-Maier EW, Barnett AS, Pokorney SD, Al-Khatib SM, Jackson KP, Bahnson TD, Ellis CR, Atwater BD, Lewis RK, et al. Left atrial appendage electrical isolation for treatment of recurrent atrial fibrillation: a meta-analysis. *JACC Clin Electrophysiol*. 2018;4:112–120. doi: 10.1016/j.jacep.2017.07.007
16. Lee RJ, Lakkireddy D, Mittal S, Ellis C, Connor JT, Saville BR, Wilber D. Percutaneous alternative to the Maze procedure for the treatment of persistent or long-standing persistent atrial fibrillation (aMAZE trial): rationale and design. *Am Heart J*. 2015;170:1184–1194. doi: 10.1016/j.ahj.2015.09.019
17. Lakkireddy DR, Wilber DJ, Mittal S, Tschopp D, Ellis CR, Rasekh A, Hounshell T, Evonich R, Chandhok S, Berger RD, et al; aMAZE Investigators. Pulmonary vein isolation with or without left atrial appendage ligation in atrial fibrillation: the aMAZE randomized clinical trial. *JAMA*. 2024;331:1099–1108. doi: 10.1001/jama.2024.3026
18. Stoltzfus JC. Logistic regression: a brief primer. *Acad Emerg Med*. 2011;18:1099–1104. doi: 10.1111/j.1553-2712.2011.01185.x
19. Bartus K, Morelli RL, Szczepanski W, Kapelak B, Sadowski J, Lee RJ. Anatomic analysis of the left atrial appendage after closure with the LARIAT device. *Circ Arrhythm Electrophysiol*. 2014;7:764–767. doi: 10.1161/CIRCEP.113.001084
20. Ellis CR, Aznaurov SG. The Left Atrial Appendage and Thromboembolic Stroke in Atrial Fibrillation: Rationale for Ligation by an Epicardial Approach: The AtriClip Stroke Trial. *J Innov Card Rhythm Manag*. 2015;6:2084–2089. doi:10.19102/icrm.2015.060602
21. Lakkireddy D, Sridhar Mahankali A, Kanmanthareddy A, Lee R, Badhwar N, Bartus K, Atkins D, Bommana S, Cheng J, Rasekh A, et al. Left atrial appendage ligation and ablation for persistent atrial fibrillation: the LAALA-AF registry. *JACC Clin Electrophysiol*. 2015;1:153–160. doi: 10.1016/j.jacep.2015.04.006
22. Han FT, Bartus K, Lakkireddy D, Rojas F, Bednarek J, Kapelak B, Bartus M, Sadowski J, Badhwar N, Earnest M, et al. The effects of LAA ligation on LAA electrical activity. *Heart Rhythm*. 2014;11:864–870. doi: 10.1016/j.hrthm.2014.01.019
23. Badhwar N, Mittal S, Rasekh A, Vasaiwala S, Musat D, P SN, Fang Q, Nentwich K, Deneke T, Chang J, et al. Conversion of persistent atrial fibrillation to sinus rhythm after LAA ligation with the LARIAT device. *Int J Cardiol*. 2016;225:120–122. doi: 10.1016/j.ijcard.2016.09.099
24. Atoui M, Pillarisetti J, Iskandar S, Earnest M, Lakkireddy D. Left atrial appendage tachycardia termination with a LARIAT suture ligation. *J Atr Fibrillation*. 2015;8:1380. doi: 10.4022/jafib.1380
25. Kawamura M, Scheinman MM, Lee RJ, Badhwar N. Left atrial appendage ligation in patients with atrial fibrillation leads to a decrease in atrial dispersion. *J Am Heart Assoc*. 2015;4:e001581. doi: 10.1161/jaha.114.001581
26. Fink T, Schlüter M, Heeger CH, Lemeš C, Maurer T, Reissmann B, Rottner L, Santoro F, Tilz RR, Alessandrini H, et al. Combination of left atrial appendage isolation and ligation to treat nonresponders of pulmonary vein isolation. *JACC Clin Electrophysiol*. 2018;4:1569–1579. doi: 10.1016/j.jacep.2018.09.022
27. Di Biase L, Burkhardt JD, Mohanty P, Sanchez J, Mohanty S, Horton R, Gallinghouse GJ, Bailey SM, Zagrodzky JD, Santangeli P, et al. Left atrial appendage. *Circulation*. 2010;122:109–118. doi: 10.1161/CIRCULATIONAHA.109.928903
28. Rillig A, Tilz RR, Lin T, Fink T, Heeger CH, Arya A, Metzner A, Mathew S, Wissner E, Makimoto H, et al. Unexpectedly high incidence of stroke and left atrial appendage thrombus formation after electrical isolation of the left atrial appendage for the treatment of atrial tachyarrhythmias. *Circ Arrhythm Electrophysiol*. 2016;9:e003461. doi: 10.1161/CIRCEP.115.003461
29. Kim YG, Shim J, Oh SK, Lee KN, Choi JI, Kim YH. Electrical isolation of the left atrial appendage increases the risk of ischemic stroke and transient ischemic attack regardless of postisolation flow velocity. *Heart Rhythm*. 2018;15:1746–1753. doi: 10.1016/j.hrthm.2018.09.012
30. Garrey WE. The nature of fibrillary contraction of the heart. Its relation to tissue mass and form. *Am J Physiol*. 1914;33:397–414. doi: 10.1152/ajplegacy.1914.33.3.397
31. Garrey WE. Auricular fibrillation. *Physiol Rev*. 1924;4:215–250. doi: 10.1152/physrev.1924.4.2.215
32. Hartley A, Shalhoub J, Ng FS, Krahn AD, Laksman Z, Andrade JG, Deyell MW, Kanagaratnam P, Sikkel MB. Size matters in atrial fibrillation: the underestimated importance of reduction of contiguous electrical mass underlying the effectiveness of catheter ablation. *Europace*. 2021;23:1698–1707. doi: 10.1093/europace/euab078
33. Lee AM, Aziz A, Didesch J, Clark KL, Schuessler RB, Damiano RJ. Importance of atrial surface area and refractory period in sustaining atrial fibrillation: testing the critical mass hypothesis. *J Thorac Cardiovasc Surg*. 2013;146:593–598. doi: 10.1016/j.jtcvs.2012.04.021
34. den Uijl DW, Delgado V, Bertini M, Tops LF, Trines SA, van de Veire NR, Zeppenfeld K, Schalij MJ, Bax JJ. Impact of left atrial fibrosis and left atrial size on the outcome of catheter ablation for atrial fibrillation. *Heart*. 2011;97:1847–1851. doi: 10.1136/hrt.2010.215335
35. Kumagai K, Toyama H. High-power, short-duration ablation during Box isolation for atrial fibrillation. *J Arrhythm*. 2020;36:899–904. doi: 10.1002/joa3.12407

36. Du W, Dai M, Wang M, Gong Q, Ye TQ, Wang H, Luo CD. Large left atrial appendage predicts the ablation outcome in hypertensive patients with atrial fibrillation. *J Electrocardiol*. 2020;63:139–144. doi: 10.1016/j.jelectrocard.2020.07.017
37. Sievert H, Rasekh A, Bartus K, Morelli RL, Fang Q, Kuropka J, Le D, Gafoor S, Heuer L, Safavi-Naeini P, et al. Left atrial appendage ligation in nonvalvular atrial fibrillation patients at high risk for embolic events with ineligibility for oral anticoagulation: initial report of clinical outcomes. *JACC Clin Electrophysiol*. 2015;1:465–474. doi: 10.1016/j.jacep.2015.08.005
38. Sabzwari SRA, Mehta NA, Garg L, Racharla L, Kurtz E, Afzal MR, Turagam MK, Lakkireddy D. Safety and efficacy of left atrial appendage closure using an epicardial suture snaring device: Systematic review and current status. *J Cardiovasc Electrophysiol*. 2021;32:2189–2198. doi: 10.1111/jce.15132
39. Tilz RR, Fink T, Bartus K, Wong T, Vogler J, Nentwich K, Panniker S, Fang Q, Piorkowski C, Liosis S, et al. A collective European experience with left atrial appendage suture ligation using the LARIAT+ device. *Europace*. 2020;22:924–931. doi: 10.1093/europace/euaa004
40. Pillai A, Kanmanthareddy A, Earnest MB, Reddy MY, Ferrell R, Nath J, Pillarisetti J, Vallakati A, Lakkireddy DR. Initial experience with post Lariat left atrial appendage leak closure with Amplatzer septal occluder device and repeat Lariat application. *Heart Rhythm*. 2014;11:1877–1883. doi: 10.1016/j.hrthm.2014.06.035
41. Cox JL, Malaisrie SC, Churyla A, Mehta C, Kruse J, Kislitsina ON, McCarthy PM. Cryosurgery for atrial fibrillation: physiologic basis for creating optimal cryolesions. *Ann Thorac Surg*. 2021;112:354–362. doi: 10.1016/j.athoracsur.2020.08.114
42. Mekhael M. Left atrial volume is the strongest predictor of atrial arrhythmia recurrence after pulmonary vein isolation alone in persistent AF: a DECAAF II substudy. *Heart Rhythm Soc*. 2023.
43. Huo Y, Gaspar T, Schönbauer R, Wójcik M, Fiedler L, Roithinger FX, Martinek M, Pürerfellner H, Kirstein B, Richter U, et al. Low-voltage myocardium-guided ablation trial of persistent atrial fibrillation. *NEJM Evid*. 2022;1:Evidoa2200141. doi: 10.1056/evidoa2200141
44. Yang G, Zheng L, Jiang C, Fan J, Liu X, Zhan X, Li J, Wang L, Yang H, Zhu W, et al; STABLE-SR-II Investigators. Circumferential pulmonary vein isolation plus low-voltage area modification in persistent atrial fibrillation: the STABLE-SR-II trial. *JACC Clin Electrophysiol*. 2022;8:882–891. doi: 10.1016/j.jacep.2022.03.012
45. DeLurgio DB, Crossen KJ, Gill J, Blauth C, Oza SR, Magnano AR, Mostovych MA, Halkos ME, Tschopp DR, Kerendi F, et al. Hybrid convergent procedure for the treatment of persistent and long-standing persistent atrial fibrillation: results of CONVERGE clinical trial. *Circ Arrhythm Electrophysiol*. 2020;13:e009288. doi: 10.1161/CIRCEP.120.009288
46. Van Der Heijden CAJ, Weberndörfer V, Vroomen M, Luermans JG, Chaldoupi S-M, Bidar E, Vernooy K, Maessen JG, Pison L, Van Kuijk SMJ, et al. Hybrid ablation versus repeated catheter ablation in persistent atrial fibrillation. *JACC Clin Electrophysiol*. 2023;9:1013. doi: 10.1016/j.jacep.2022.12.011
47. Doll N, Weimar T, Kosior DA, Bulava A, Mokracek A, Mönnig G, Sahu J, Hunter S, Wijffels M, Van Putte B, et al. Efficacy and safety of hybrid epicardial and endocardial ablation versus endocardial ablation in patients with persistent and longstanding persistent atrial fibrillation: a randomised, controlled trial. *eClinicalMedicine*. 2023;61:102052. doi: 10.1016/j.eclinm.2023.102052